

SEJAL SHANKAR DESAI

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PROFESSIONAL SUMMARY

Results-driven Automation and Programming Specialist with experience in laboratory robotics, full-stack web development, and process optimization. Proven track record of designing and deploying Hamilton liquid handling workflows, Python-based automation tools, and scalable web applications. Adept at cross-functional collaboration, Agile methodologies, and applying Lean Six Sigma principles to drive efficiency and quality in regulated lab environments.

EDUCATION

Northeastern University, Boston, MA
Master of Science in Information Systems

Sept 2022 – May 2024

University of Mumbai, India
Bachelor of Engineering in Electronics

July 2018 – June 2022

EXPERIENCE

Eurofins | Kalamazoo, MI (Providing Service to Zoetis)
Automation and Programming Specialist

Nov 2024 – Present

- Spearheaded a cell line development project on Hamilton 2, utilizing Twister channels to automate cap removal and re-capping of 50 mL conical tubes, enabling consistent and repeatable high-throughput sample preparation workflows.
- Engineered a Python-based Hamilton method file extraction tool integrated with an AI model to parse, analyze, and document instrument code logic and execution flow, reducing method review time and accelerating onboarding.
- Optimized Andrew+ pipetting robot experiment runtime by 75% (60 min → 15 min) through method redesign, deck layout optimization, and parallelization of liquid-handling operations, increasing throughput and reproducibility.
- Developed Excel-driven input templates for the glycan assay on Andrew+, streamlining parameter entry, volume mapping, and plate configuration to reduce setup time and minimize manual entry errors.
- Enhanced existing lab automation applications using VB.NET, improving functionality, stability, and usability of tools across active production workflows.
- Designed and validated Hamilton VENUS protocols for the parasitology group, including a flea method requiring precise 2 µL dispense accuracy.
- Collaborated cross-functionally with scientists, QA, and lab operations to align automation protocols with compliance standards, improving assay reproducibility and consistency.

Zoetis | Kalamazoo, MI
Web Developer Intern

June 2024 – Nov 2024

- Architected and delivered a full-stack lab inventory management system using React 18 and .NET Core 8, centralizing consumable tracking across multiple lab locations and reducing lookup time by 70%.
- Implemented JWT-based authentication and role-based access control (Admin/User) with RESTful API endpoints supporting full CRUD operations for inventory and order management.
- Engineered a responsive UI using Styled Components and React Hooks with dark/light mode, real-time auto-refresh, and mobile-first design, improving cross-device accessibility.
- Utilized an existing SQL Server database, writing optimized queries via Entity Framework Core with indexed columns to improve application performance and data retrieval speed.
- Built custom middleware for request validation, exception tracking, and graceful degradation, achieving 99.5% application uptime and seamless recovery across network failures.
- Streamlined order approval workflows and inventory lookup through an admin dashboard with real-time status updates, reducing manual overhead and improving operational efficiency.

TCR Innovation | Mumbai, India
Front-End Web Developer

Sept 2021 – Feb 2022

- Partnered with UX/UI designers, back-end developers, and stakeholders to define requirements and deliver responsive, user-centric web applications aligned with business objectives.
- Built and maintained client-side applications using HTML5, CSS3, JavaScript, and jQuery, ensuring cross-browser compatibility, accessibility, and optimized page performance.
- Applied component reusability and performance optimization techniques, resulting in measurable improvements in user engagement and overall application quality.

TECHNICAL SKILLS

Languages:	Python, Java, C, C++, Swift, VB.NET
Web Development:	HTML5, CSS3, SCSS, JavaScript, Bootstrap, ReactJS, Tailwind CSS, Node.js, ExpressJS, OpenAPI
Databases:	MySQL, SQL Server, MongoDB, Core Data
Automation:	Hamilton Liquid Handlers, Andrew+, Agilent Bravo, Hive Robot, Tempest, Fanuc Robotic Arm
Lab Software:	VENUS, Green Button Go, One Lab Waters
Design Tools:	Figma, Adobe XD, Sketch, Balsamiq, Axure, Moqup
Project Tools:	Jira, Git, Agile/Scrum

CERTIFICATIONS

- Hamilton VENUS Software Training — Hamilton Company
- Bionex Green Button Go Software Training — Bionex Instruments
- Six Sigma Yellow Belt Certification — Lean Six Sigma Institute